



1M1B AI FOR SUSTAINABILITY

EcoSort AI

An Agentic AI Assistant for
Intelligent Waste Segregation and Responsible Disposal

Multimodal AI • RAG • LangGraph • IBM watsonx.ai • IBM BOB

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AICTE STU ID • STU69f200fb9d16a1777467643

12

PRIMARY SDG

Responsible
Consumption

11

secondary • Sustainable Cities

1M1B × IBM SkillsBuild × AICTE

The Problem EcoSort AI addresses

Waste decisions often happen in seconds—while the right handling route can depend on the material, risk, and local collection practices.

How might we use multimodal AI to identify household waste and provide context-grounded, safety-conscious segregation and disposal guidance so that responsible consumption becomes easier at the point of disposal?

WHAT SUCCESS LOOKS LIKE

● Clarity

Explain what the item is and why it was classified.

● Grounding

Use retrieved waste guidance as the factual basis.

● Safety

Avoid dangerous instructions and refuse to guess when uncertain.

● Local awareness

Remind users that collection rules can differ by locality.

THE PAIN POINT

“What bin / route does this item belong to?”

WHY AI?

- Classify text and image inputs quickly
- Retrieve relevant guidance instead of guessing
- Route uncertain / special-handling cases safely

Design principle: build a decision-support assistant—not an authority that pretends local rules are universal.

EcoSort AI at a glance

A multimodal, agentic decision-support prototype that combines foundation models with a lightweight retrieval layer.

INPUT

TEXT

IMAGE

LOCATION

One simple user journey

Describe the waste item, optionally upload a photo, and provide a locality when available.

1

Identify

2

Retrieve

3

Reason

4

Recommend

AGENTIC + MULTIMODAL

The AI pipeline keeps classification, retrieval, routing, and recommendation as explicit stages.

PRIMARY SDG

12

Responsible Consumption and Production

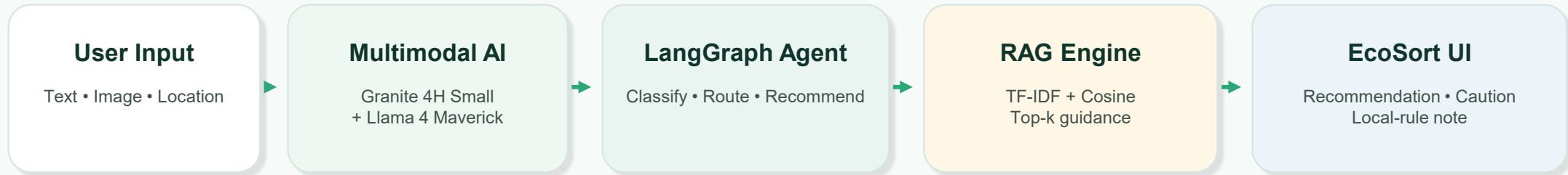
TARGET USERS

- Students & campus communities
- Households & everyday consumers
- Sustainability educators / awareness programs

Secondary alignment • SDG 11 — Sustainable Cities and Communities

System Architecture: From User input to Grounded Guidance

The Prototype deliberately separates Model Inference, Orchestration, Retrieval and Presentation.



What happens inside the agent?

01

Classify

Allowed categories + uncertainty

02

Retrieve

Build query → top 3 guidance

03

Route

Standard / special / uncertain

04

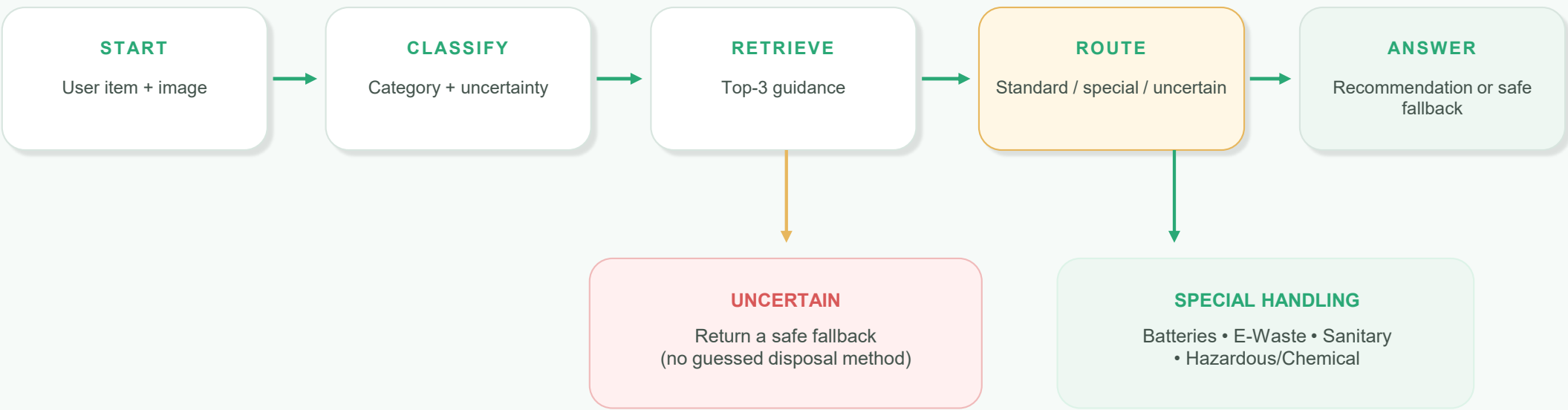
Recommend

Grounded, safety-conscious answer

Key design choice: the model does not directly invent a disposal route; it is given retrieved guidance and explicit safety constraints.

LangGraph Workflow & Routing Logic

The agent is implemented as an explicit state graph, making the reasoning path inspectable and easy to extend.



Allowed waste categories

Organic / Food	Paper / Cardboard	Plastic	Glass	Metal
E-Waste	Batteries	Sanitary / Hygiene	Hazardous / Chemical	Mixed / Unknown

Two Model Roles, One Sustainability Workflow

The Project uses a text-capable IBM Granite model for reasoning and a multimodal model available in the project environment for image understanding.

TEXT + REASONING

IBM Granite 4H Small

Used for:

- Waste classification from structured prompts
- Interpreting retrieved guidance
- Generating concise recommendations and cautions

Model strategy: prompt engineering + grounding, not custom fine-tuning.

IMAGE UNDERSTANDING

Llama 4 Maverick

Multimodal path

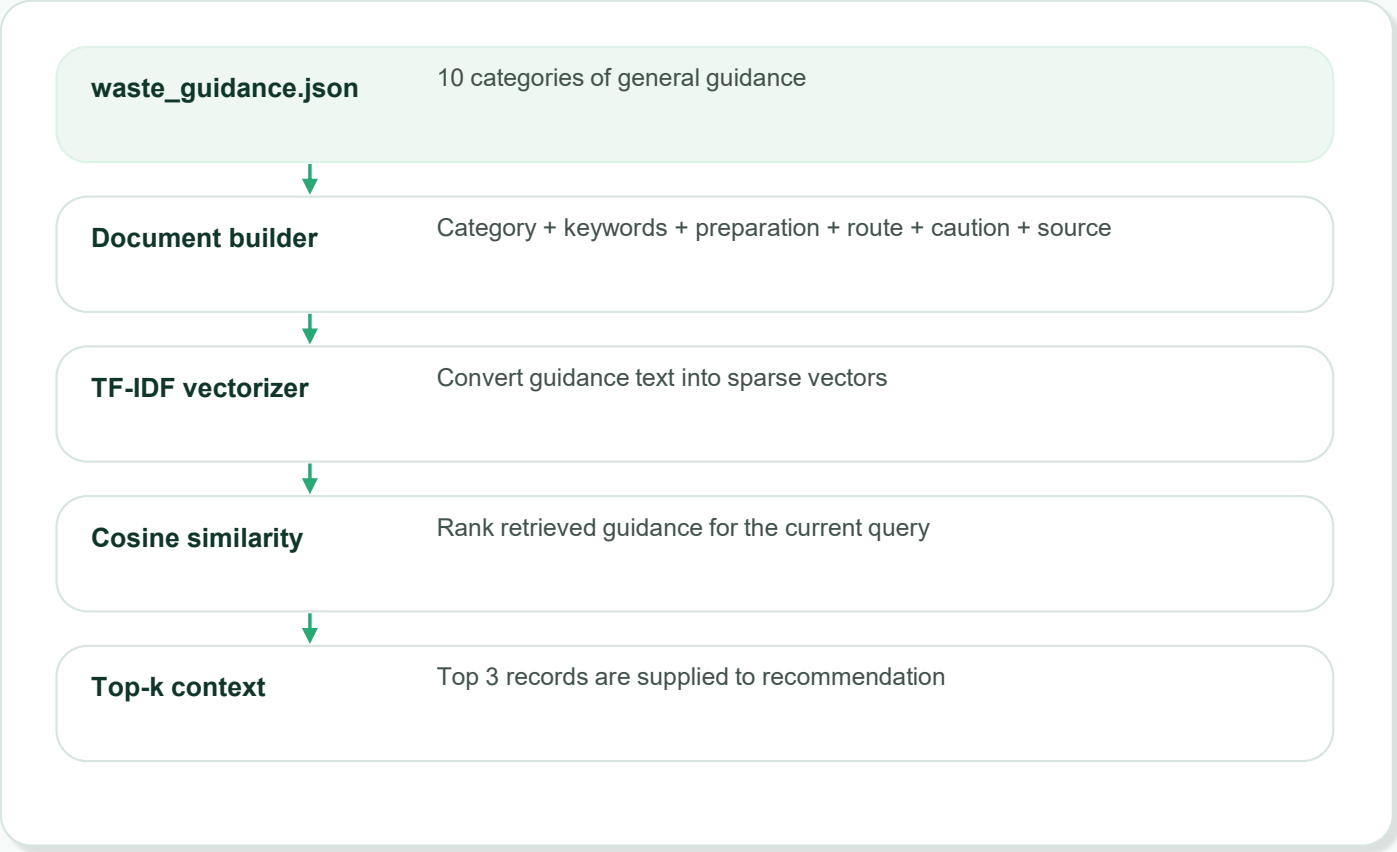
- Analyzes uploaded waste images
- Provides a visual description / clues
- Feeds the description into the same classification → RAG path

Vision model is selected from the models supported by the project environment.

One shared downstream path **Text description / image description → classify → retrieve guidance → safe recommendation**

The Knowledge Layer: Grounded Guidance, not Black-box Answers

A lightweight retrieval component turns a curated JSON knowledge base into context for the recommendation model.



WHY THIS MATTERS

The recommendation model is given retrieved evidence before it answers.

CURRENT KNOWLEDGE SCOPE

General educational / safety guidance with an explicit “verify local rules” note.

Known limitation

TF-IDF can miss semantic synonyms; future embeddings can improve recall.

RAG similarity scores shown in testing are retrieval scores—not model confidence probabilities.

Safety and responsible-use boundaries are built into the workflow

The internship guide explicitly requires fairness, transparency, ethics, and privacy considerations.

1

Transparency

Explain the detected item, category, reason, and retrieved guidance.

BUILT-IN

2

Safety

Special handling for batteries, e-waste, sanitary and hazardous categories.

BUILT-IN

3

Uncertainty

If evidence is insufficient, use Mixed / Unknown instead of guessing.

FAIL-SAFE

4

Privacy

Keep API keys in environment/secrets; avoid exposing private data.

BUILT-IN

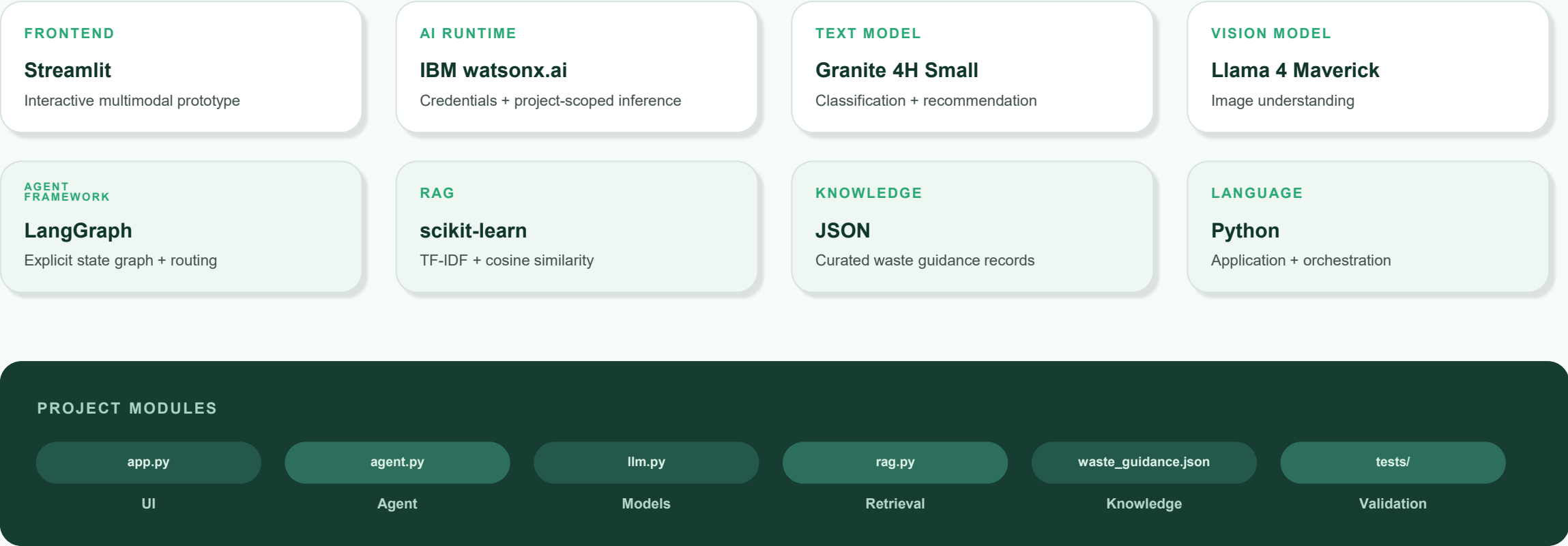
Core rule

EcoSort AI is a decision-support prototype. Local authorities / authorized collection providers remain the source of truth for actual disposal requirements.

Aligned with the internship requirement for Responsible AI Considerations.

Implementation Stack & Project Structure

The prototype is intentionally modular so each major responsibility can be demonstrated independently.



No private credentials are included in the project files; runtime secrets are loaded from environment / Streamlit secrets.



EcoSort AI

🌱 SDG 12 • SDG 11

Intelligent Waste Segregation & Responsible Disposal Assistant

IBM watsonx.ai

Agentic AI

RAG

Multimodal



Describe a waste item or upload an image. EcoSort AI combines AI classification with retrieved waste guidance. Local disposal rules can vary, so verify with the appropriate local authority or authorized collection service.

Try an example

Empty plastic water bottle

Used lithium battery from a power bank

Banana peel

What do you want to identify?

Example: Used AA battery from a TV remote

Upload a waste image



Upload

200MB per file • JPG, PNG, WEBP



Location

Example: Kolkata, West Bengal, India



Analysis mode

Using text-based analysis

 Analyze Waste New

 Upload a waste image



Screen...ng.com.jpeg ×
225.5KB



Image ready for analysis

 Analyze Waste

 New

✓  Image analysis complete

✓  EcoSort analysis complete

AI Result

 Location

Example: Kolkata, West Bengal, India

 Analysis mode

Using image-based analysis

Detected item

smartphone

Waste category

 E-Waste

RAG knowledge match

0.40

✓ The item was classified with sufficient evidence for a practical recommendation.

✓ What to do

Detected item: smartphone

Waste category: E-Waste (electronic waste)

What to do:

- Keep the smartphone separate from ordinary household waste.
- Locate an e-waste collection point or recycling channel in your area (e.g., municipal e-waste drop-off sites, designated recycling bins, or retailer take-back programs).
- Follow the instructions provided at the collection point for safely depositing the device.

Important caution:

- Do **not** dismantle the smartphone yourself just to dispose of it.
- If the device contains a removable battery, handle the battery separately according to local battery-recycling guidelines—many regions require batteries to be recycled apart from the rest of the e-waste.

Local-rule note:

- Collection options for e-waste can vary by municipality. Verify the specific e-waste recycling or collection services available in your locality before disposing of the smartphone. If you are unsure, contact your local waste-management authority or visit their website for the most up-to-date guidance.

▼ 🔍 Why did EcoSort AI classify it this way?

The item is clearly an electronic device, specifically a smartphone, which falls under the category of electronic waste (e-waste).

▼ 🖼️ Image understanding result

The image shows a white smartphone with a cracked screen lying on a dark wooden surface. The phone's screen is extensively cracked, with multiple lines radiating from the center of the impact area.

Visible Object:

- The object is a smartphone.
- It is white in color.
- The phone has a cracked screen.

Material:

- The phone's body appears to be made of plastic or metal, but the exact material cannot be determined from the image.
- The screen is made of glass, as indicated by the cracking pattern.

Useful Clues for Waste Classification:

- The phone is electronic waste (e-waste) due to its nature as an electronic device.
- The cracked screen suggests that the phone is broken and likely non-functional or not worth repairing, making it a candidate for disposal.
- The presence of a battery (not visible but inferred as it is a smartphone) indicates that it contains hazardous materials that require proper disposal.
- The phone's components, such as the circuit board, metals, and other internal parts, are recyclable and should be handled accordingly.

Overall, the image suggests that the smartphone is waste that needs to be disposed of properly, preferably through an e-waste recycling program to handle its hazardous and recyclable components.

▼ Knowledge used (3 retrieved records)**Record 1**

Category: E-Waste Keywords: phone laptop charger keyboard mouse computer electronics circuit Preparation: Keep electronic waste separate from ordinary household waste. Route: Use an appropriate e-waste collection or recycling channel available in the locality. Caution: Do not dismantle electronic equipment yourself just to dispose of it. Source: General e-waste guidance; verify current local collection arrangements

RAG similarity: 0.397

Record 2

Category: Hazardous / Chemical Waste Keywords: paint pesticide chemical solvent cleaner oil chemical container Preparation: Keep potentially hazardous chemical material separate from ordinary household waste. Route: Follow the local hazardous-waste or authorized collection instructions. Caution: Do not mix unknown chemicals or pour hazardous substances into drains. Source: General safety guidance; verify local authority instructions

RAG similarity: 0.204

Record 3

Category: Glass Keywords: glass bottle jar glass container Preparation: Keep glass separate from loose organic waste and handle broken glass carefully. Route: Use the local glass or recyclable collection channel where available. Caution: Broken glass can cause injury. Handle it carefully. Source: General educational guidance

RAG similarity: 0.188

Record 2

Category: Hazardous / Chemical Waste Keywords: paint pesticide chemical solvent cleaner oil chemical container Preparation: Keep potentially hazardous chemical material separate from ordinary household waste. Route: Follow the local hazardous-waste or authorized collection instructions. Caution: Do not mix unknown chemicals or pour hazardous substances into drains. Source: General safety guidance; verify local authority instructions

RAG similarity: 0.204

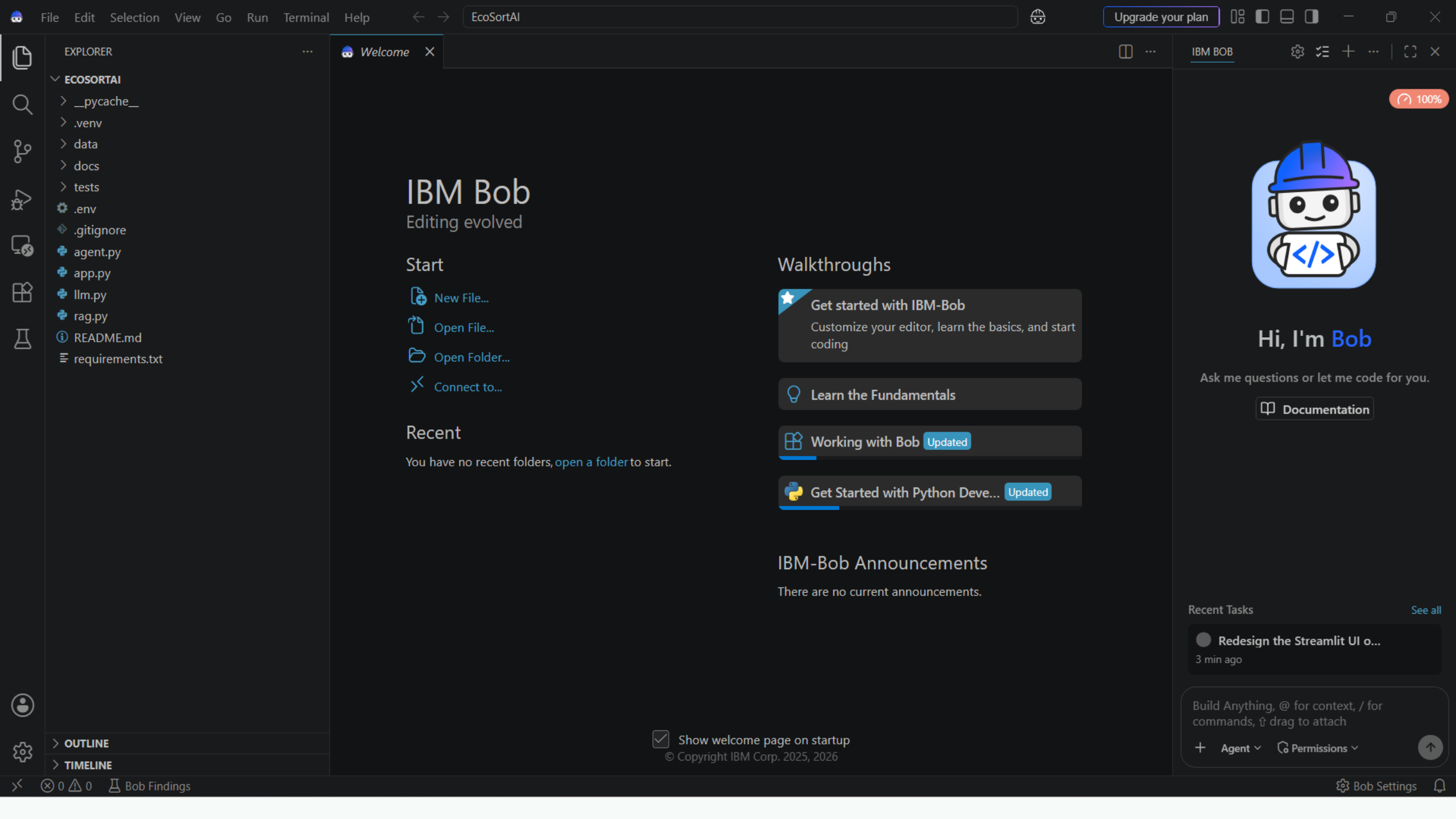
Record 3

Category: Glass Keywords: glass bottle jar glass container Preparation: Keep glass separate from loose organic waste and handle broken glass carefully. Route: Use the local glass or recyclable collection channel where available. Caution: Broken glass can cause injury. Handle it carefully. Source: General educational guidance

RAG similarity: 0.188

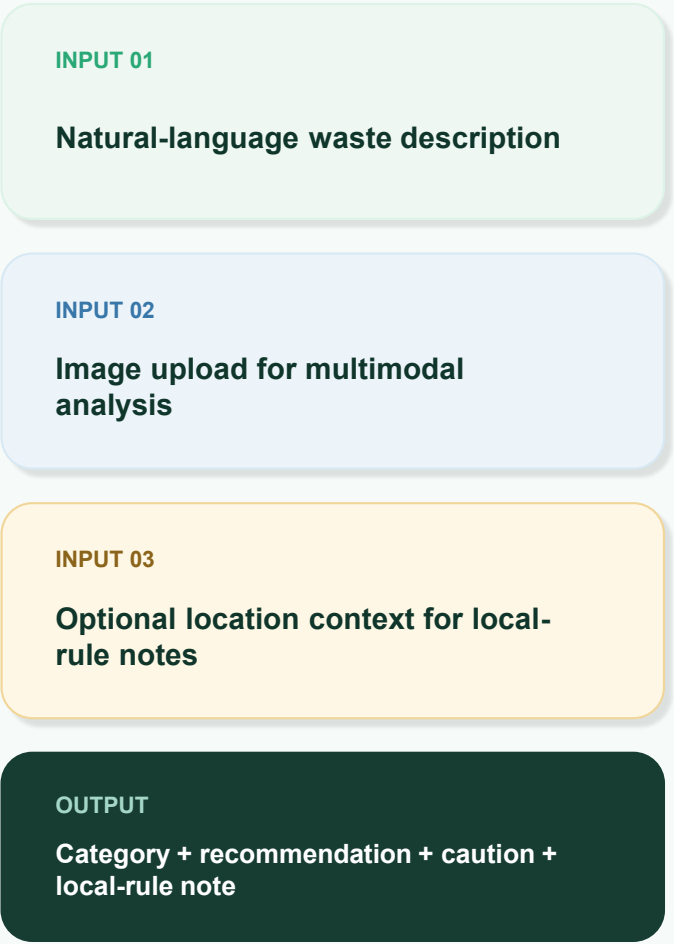
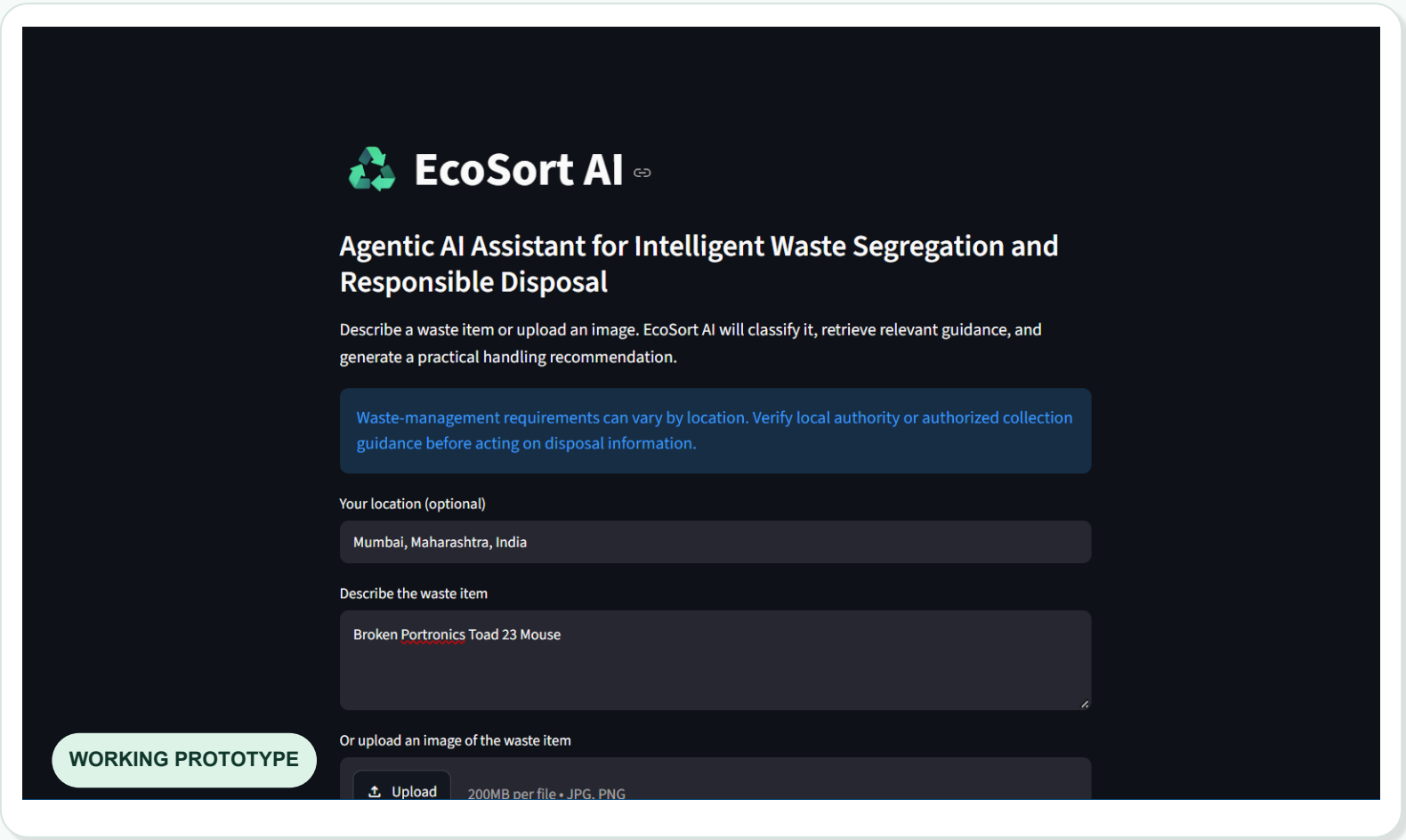
⌵ ⚙️ How EcoSort AI works

1. **Input** → text description and/or waste image
2. **AI understanding** → text and image-capable foundation models
3. **Agent workflow** → classification and decision logic
4. **RAG** → relevant waste-management guidance is retrieved
5. **Recommendation** → practical handling and disposal guidance is generated



Prototype UI: text + image + location in one workflow

The working Streamlit prototype accepts natural-language descriptions, optional images, and an optional locality.



Prototype outputs: grounded recommendations in action

The following result snapshot comes from the tested prototype workflow.

ECOSORT AI Recommendation

Detected item: Destroyed Mobile Phone
Waste category: E-Waste

What to do:

- Keep the destroyed mobile phone separate from ordinary household waste.
- Locate an appropriate e-waste collection or recycling channel in your locality and deposit the phone there.

Important caution:

- Do not attempt to dismantle the phone yourself for disposal.
- Avoid puncturing, burning, crushing, or otherwise damaging the device before recycling.

Local-rule note:

- Verify current local collection arrangements, as specific drop-off points or collection schedules may vary by region.

AI Classification

Detected item	Waste category
IMAGE-ASSISTED E-WASTE DEMO	E-Waste

DETECTED

Destroyed Mobile Phone • E-Waste

SAFETY MESSAGE

Do not dismantle or further damage the device for disposal.

GROUNDING OUTPUT

The recommendation explicitly separates E-Waste from ordinary household waste and defers to local collection arrangements.

Prototype evidence • local Streamlit run

Observed test cases & retrieval behavior

Validation focused on classification, RAG retrieval, uncertainty handling, and practical recommendation generation.

TEST INPUT	CATEGORY	RAG TOP SCORE	OBSERVED BEHAVIOR
Empty plastic water bottle	Plastic	0.766	Correct category; recommendation generated
Used lithium battery	Batteries	0.400	Separate route + safety caution
Old laptop	E-Waste	0.186	Correct special-handling category
Cardboard box	Paper / Cardboard	0.337	Dry recyclable route retrieved
Unknown powder	Mixed / Unknown	0.415	uncertainty=True; safe fallback

Validation takeaway

Correct categories + grounded responses + explicit uncertainty path.

Important

RAG top scores are TF-IDF/cosine similarity, not probability/confidence.

Current limitations → next-generation roadmap

The prototype is intentionally honest about what it knows, what it does not know, and where engineering can improve it.

CURRENT LIMITATIONS

- TF-IDF retrieval is lexical; paraphrases may retrieve weaker matches.
- Knowledge is general guidance, not a verified municipal rule database.
- Image results depend on the vision model available in the watsonx environment.
- No measured environmental impact yet; the current stage is a working prototype.

The prototype already exposes these boundaries to users through uncertainty and local-rule messaging.

FUTURE SCOPE

01 Semantic embeddings + vector store

Improve retrieval recall

02 Verified municipal datasets

Add locality-specific policy context

03 Regional languages

Make guidance more accessible

04 Feedback + analytics

Learn from corrected classifications

05 Sustainability Hub

Add Water Assistant as a second module

Water module concept shared IBM watsonx + agent layer, separate water knowledge base

Expected value of the prototype

The value proposition is practical: better point-of-disposal guidance, safer handling, and a reusable AI + sustainability architecture.



Safer handling

Special categories surface explicit cautions before disposal.



Better segregation

Users receive a category + preparation + route recommendation.



More transparent AI

Classification reason + retrieved guidance make the result inspectable.



Reusable architecture

The same agent/RAG pattern can extend to water and other SDG domains.

SUBMISSION SNAPSHOT

Working Streamlit prototype

IBM watsonx.ai

Granite + multimodal model

LangGraph agent

RAG knowledge layer

Text + image validation

Responsible AI boundaries

No numerical environmental impact is claimed yet; impact is stated as expected until field data / user studies are collected.



THANK YOU

EcoSort AI

**An Agentic AI Assistant for
Intelligent Waste Segregation and Responsible
Disposal**

A multimodal AI + RAG prototype aligned primarily with SDG 12.

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GitHub Repository : <https://github.com/zeeshanalisk/EcoSort-AI>
Project Deployment link : <https://ecosort-ai-sdg.streamlit.app/>
Public Demo URL : [Video Link, Click Here](https://drive.google.com/file/d/1Drb6H13LtF0_jc5B_c456_iqfWzh0h8p/view?usp=sharing)
https://drive.google.com/file/d/1Drb6H13LtF0_jc5B_c456_iqfWzh0h8p/view?usp=sharing

12

RESPONSIBLE
CONSUMPTION

11

SUSTAINABLE
CITIES